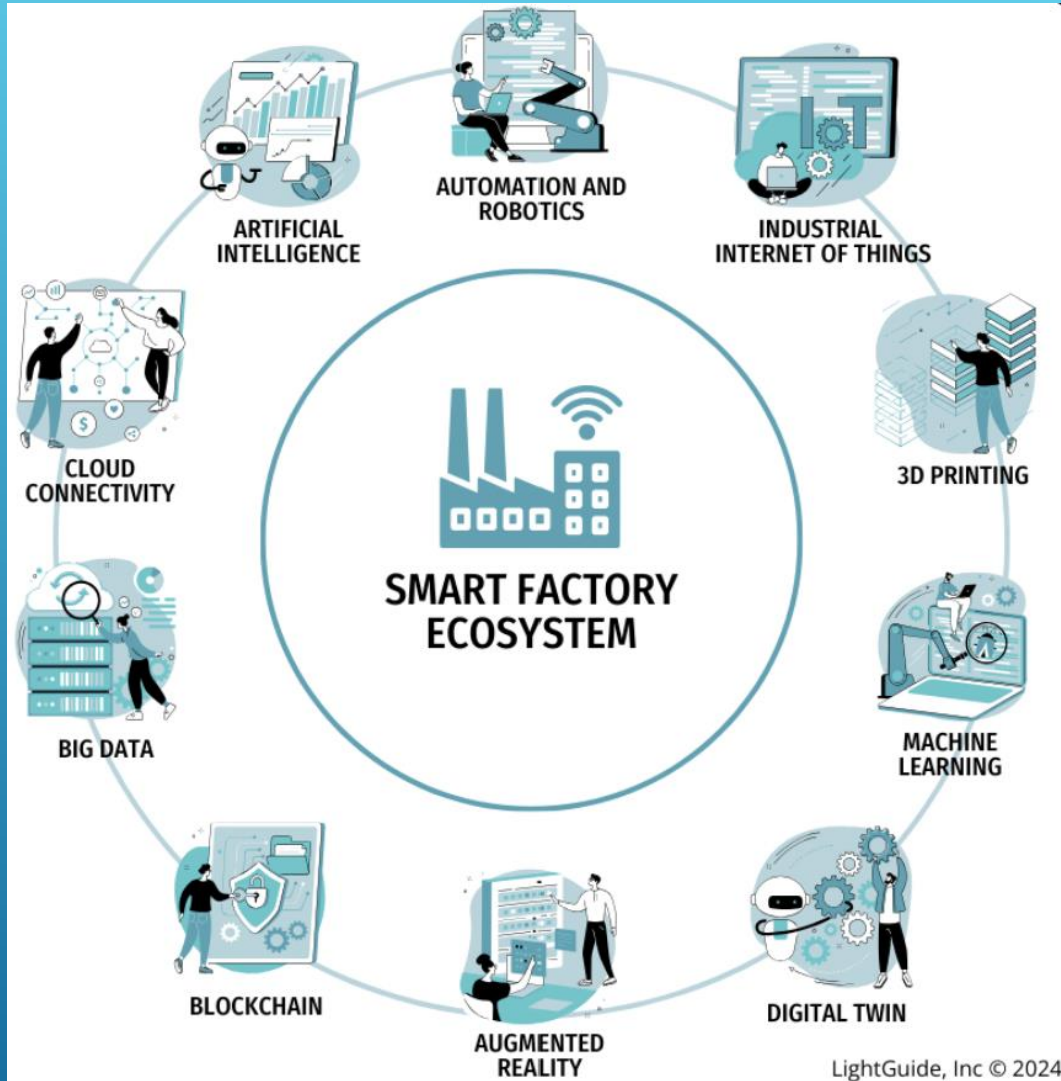


IOT IOMT IIOT IN 2025 : MASTERING MASSIVE DATA CHALLENGES WITH INDUSTRIAL PROTOCOLS.

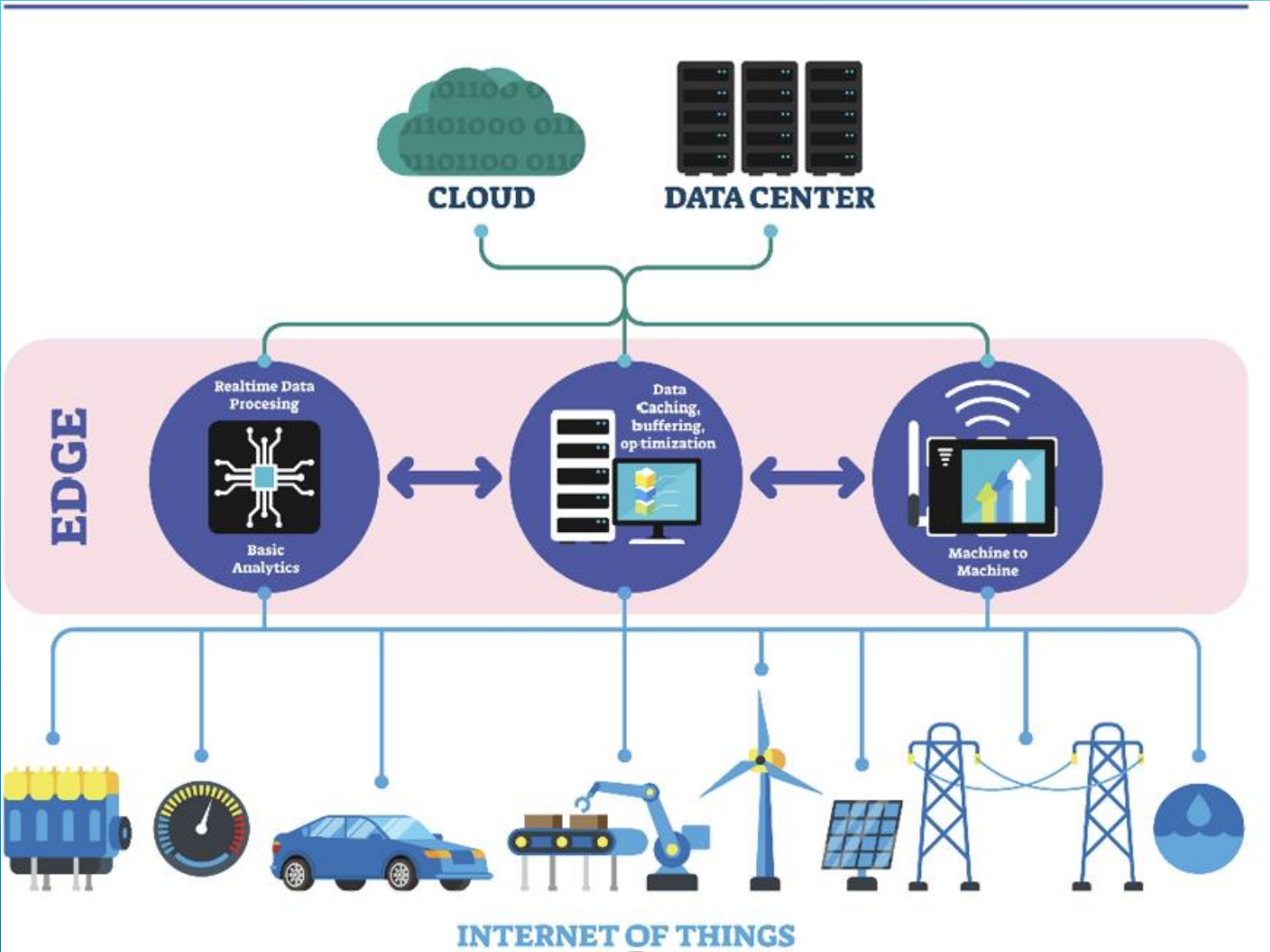
By Khaoula Ben Ali Ph.D in Electrical Engineering
and Telecommunications

SECTION 1: FROM THE SMART FACTORY TO BIG DATA



WHAT IS A SMART FACTORY?

- A highly digitized and connected production facility.
- Uses sensors, machines, and IIoT to collect data in real time.
- Integrates OT (factory floor control) with IT (enterprise/cloud analytics).
- Enables predictive maintenance, efficiency, and flexibility.
- Transforms raw signals into actionable insights for decision-making.



3 KEY TALKING POINTS AND TWO OBJECTIVES

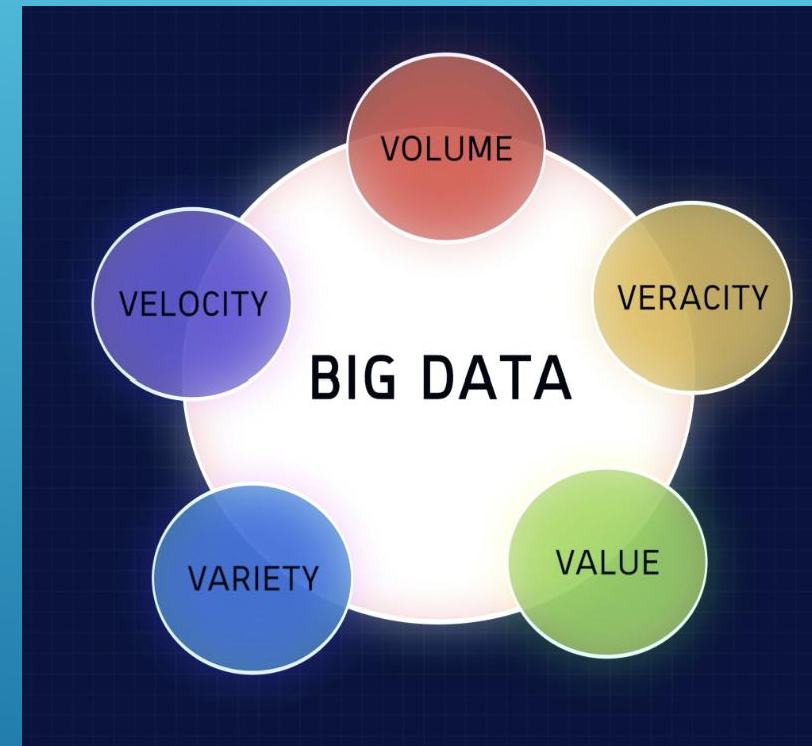
I. 3 Key Talking Points

1. The Volume and Velocity Challenge (Big Data)
2. The Role of the Edge and Immediate Analytics
3. The Link to Protocols



BIG DATA 3/ 4 5V TO
7V(VALIDITY AND
VOLATILITY)

Unit Name	Abbreviation	Multiplier (SI)	Size in Bytes
Kilobyte	KB	10^3	1,000 Bytes
Megabyte	MB	10^6	1,000,000 Bytes
Gigabyte	GB	10^9	1,000,000,000 Bytes
Terabyte	TB	$\mathbf{10^{12}}$	1 Trillion Bytes
Petabyte	PB	$\mathbf{10^{15}}$	1,000 Terabytes
Exabyte	EB	10^{18}	1,000 Petabytes
Zettabyte	ZB	10^{21}	1,000 Exabytes



THE LINK TO PROTOCOLS

WHY specific protocols are necessary to move data efficiently across the IIoT architecture?

- ▶ **The Necessity of the Protocol Translator (Bridging the Gap)**

Characteristic	OT (Factory Floor, Machines)	IT (Cloud, Enterprise)	The Gap
Priority	Safety, Reliability, Real-time (Low Latency)	Data Security, Confidentiality, Scalability	Different priorities.
Protocols	Modbus, Profinet (Often proprietary, low native security)	HTTP, TCP/IP(not suited for real-time)	Different communication languages.
Data Context	Raw value (E.g.: Register 40001 = 85.5)	Object model (E.g.: Machine X - Motor Temperature = 85.5 °C)	IT doesn't understand the raw data meaning.

THE LINK TO PROTOCOLS

WHY specific protocols are necessary to move data efficiently across the IIoT architecture?

- ▶ **The Role of the Bridge (OPC UA)**
- ▶ This is why the **OPC UA** protocol is so important:
- ▶ **It was born from IT principles:** It uses the IT architecture (TCP/IP) and has **encryption (TLS) built-in natively** to meet the security requirements of the IT domain.
- ▶ **It speaks to OT:** It is capable of taking unencrypted data from protocols like **Modbus (OT)** and encapsulating it within a **secure, encrypted OPC UA connection** for transfer to the enterprise network (IT).

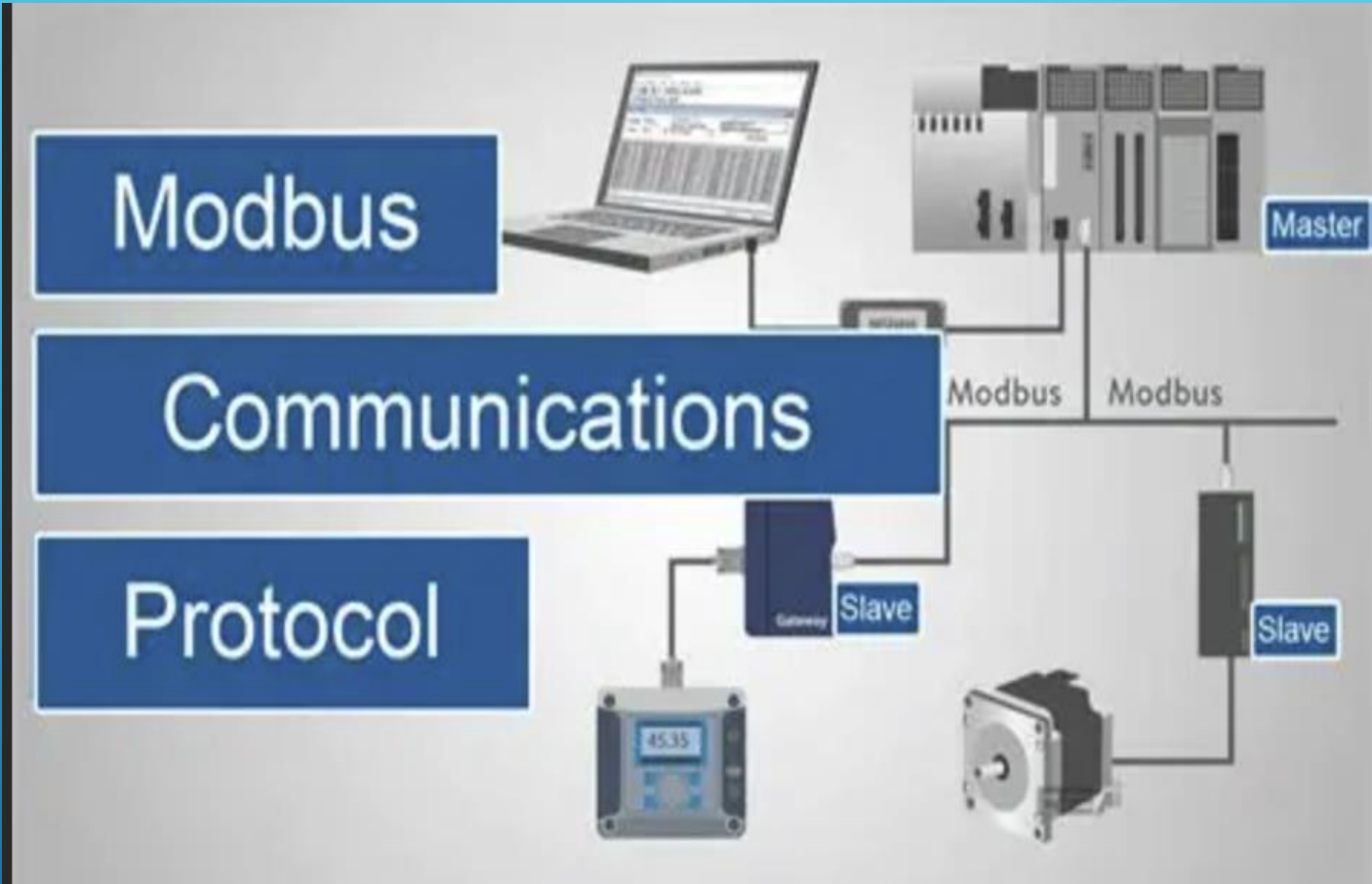
THE LINK TO PROTOCOLS

OT/IT Gap

SECTION 2: THE ENDURING UTILITY OF MODBUS

Despite the OT/IT Gap

- ▶ Q1: So, why does the industry continue to use Modbus, such a simple and unsecured protocol, inside the core Industrial Controllers?



🔌 MODBUS PROTOCOL: DEFINITION AND ENDURING UTILITY

- **Primary Role:**

Modbus is the **foundational protocol for low-level communication inside industrial devices.**

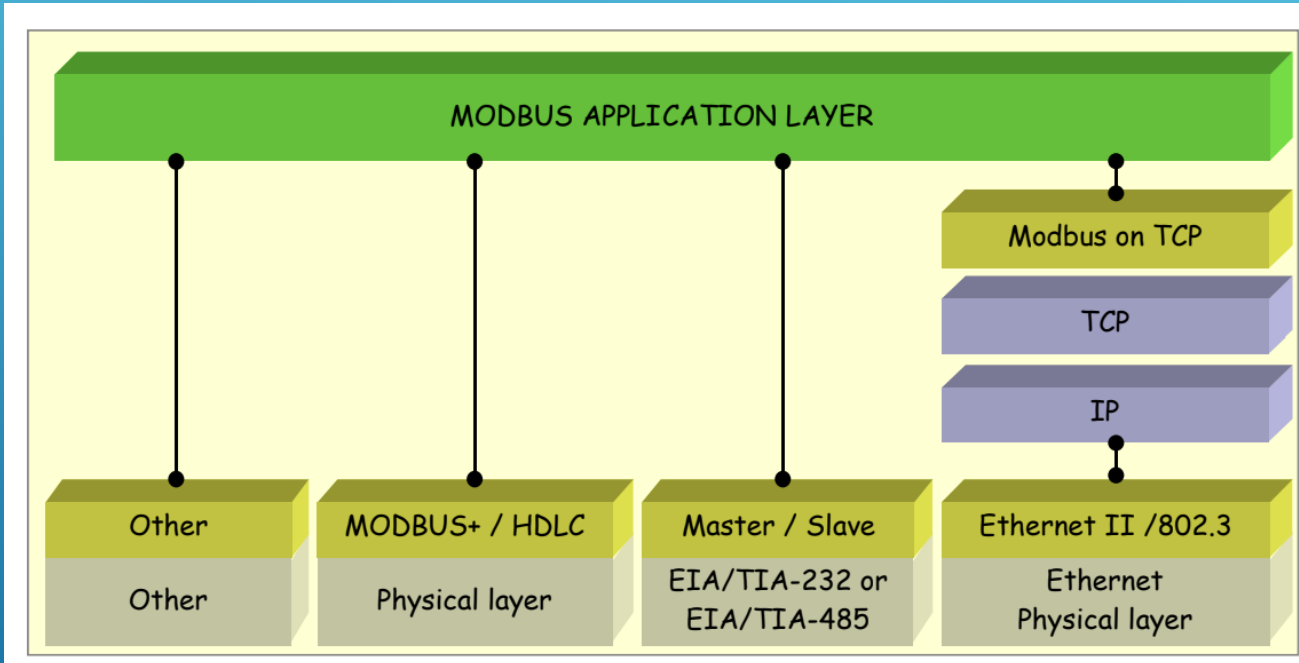
- **Historical Model:**

It operates on a **Master/Slave** model, where a single Master (often a PLC) queries a Slave device (e.g., a smart wattmeter) for data.

- **Users:** Companies like **Schneider Electric** and virtually all manufacturers of industrial controllers continue to rely on it.

MODBUS: LAYERS MODEL

Modbus Plus was developed to overcome the speed and capacity limitations of serial Modbus (RTU) while retaining the **deterministic control** required in OT environments. The **token passing** mechanism ensures that every device gets guaranteed access to the network, making it highly reliable for control loops.



- ▶ Response: because Modbus remains fundamental to the OT world. Its enduring utility comes down to three main reasons tied to its design philosophy:
- ▶ **1. Simplicity and Ubiquity (Low Complexity, High Adoption)**
- ▶ **2. Speed and Predictability (Deterministic Timing)**
- ▶ **3. Ease of Diagnostics**

THE ENDURING UTILITY OF MODBUS

There are **many variants of Modbus**. The organization responsible for its maintenance and standardization (the **Modbus Organization**) has written new formats and protocols for its continued use and relevance in modern industrial environments.

SECTION 3: LET'S BUILD A SCENARIO

- ▶ **Key Players and Platforms**
- ▶ **Siemens (MindSphere)**
- ▶ **Siemens** is a global powerhouse in industrial automation, manufacturing, and technology.
- ▶ **MindSphere** is their specific **Industrial IoT (IIoT) operating system** and cloud platform. It's designed to connect industrial equipment (like their own PLCs and other devices) and machinery to the cloud to collect and analyze data. * It focuses on providing solutions for **predictive maintenance, energy management, and resource optimization** within the industrial sector.

Q2: Who uses these protocols and **why is Modbus needed?**

Protocol	Primary Role in IIoT	Enterprise User / Case Study	Contextual Link
OPC UA	Data modeling, security, and interoperability	Siemens (MindSphere) / Microsoft	OPC UA is the cornerstone for defining standardized data models across different manufacturers. Siemens uses it to integrate factory data into its MindSphere platform for analytics.
MQTT	Lightweight, efficient, and scalable data transport from the Edge to the Cloud (essential for Big Data ingestion).	Amazon Web Services (AWS) / Azure	AWS uses MQTT as the primary ingestion protocol for AWS IoT Core , allowing millions of devices to stream data efficiently to their massive storage and analysis platforms.
Modbus	The foundational protocol for low-level communication <i>inside</i> industrial devices (PLCs, motor controllers, sensors).	Schneider Electric	Modbus is the historical base for control systems. It ensures the simple, predictable, and fast local control required for machine safety and performance.

LET'S BUILD A SCENARIO

Key Players and Platforms

Siemens (MindSphere)

- ▶ **Context: Motor Monitoring**
- ▶ **Master:** The Programmable Logic Controller (PLC) that controls a production line.
- ▶ **Slave:** The metering device (a smart wattmeter) connected to Motor M-42.
- ▶ **Objective:** The PLC needs to read the instantaneous power consumed by the motor.

SCENARIO: MODBUS DATA REQUEST (MASTER TO SLAVE)

To illustrate the direct function of **Modbus** (the Master-Slave protocol), we will focus solely on the data request between the Master and the Slave, without the IIoT integration layer. This is the essence of **OT** (Operational Technology) communication.

- ▶ **Target Data:** Power of Motor M-42.
- ▶ **Modbus Address:** Input Register **30010**.
- ▶ **Slave Address:** The device has a Slave ID of **4**.

STEP 1: THE MASTER SEEKS TO READ

The instantaneous power is stored in a register on the Slave device. In the Modbus standard, reading measurement data is done via **Input Registers**, typically starting from the functional address 30001.

- ▶ **A work to do: Request Construction (Master's Message).**
- ▶ The Modbus Appendix shows how to construct the frames

STEP 2: REQUEST CONSTRUCTION (MASTER'S MESSAGE)

The PLC (Master) creates and sends a Modbus "frame" (via Modbus TCP or RTU). This frame must contain the following information:

Position	Field	Value (Decimal)	Value (Hexadecimal)	Role in our scenario
Byte 1	Slave Address	4	04	ID of the M-42 wattmeter.
Byte 2	Function Code	4	04	Based on the Appendix (Read Input Register).
Bytes 3-4	Starting Address	9	00 09	Internal address of register 30010 (30010 - 30001 = 9).
Bytes 5-6	Number of Registers	1	00 01	We only want to read one register.

STEP 3: CONSTRUCTION (MASTER'S MESSAGE)

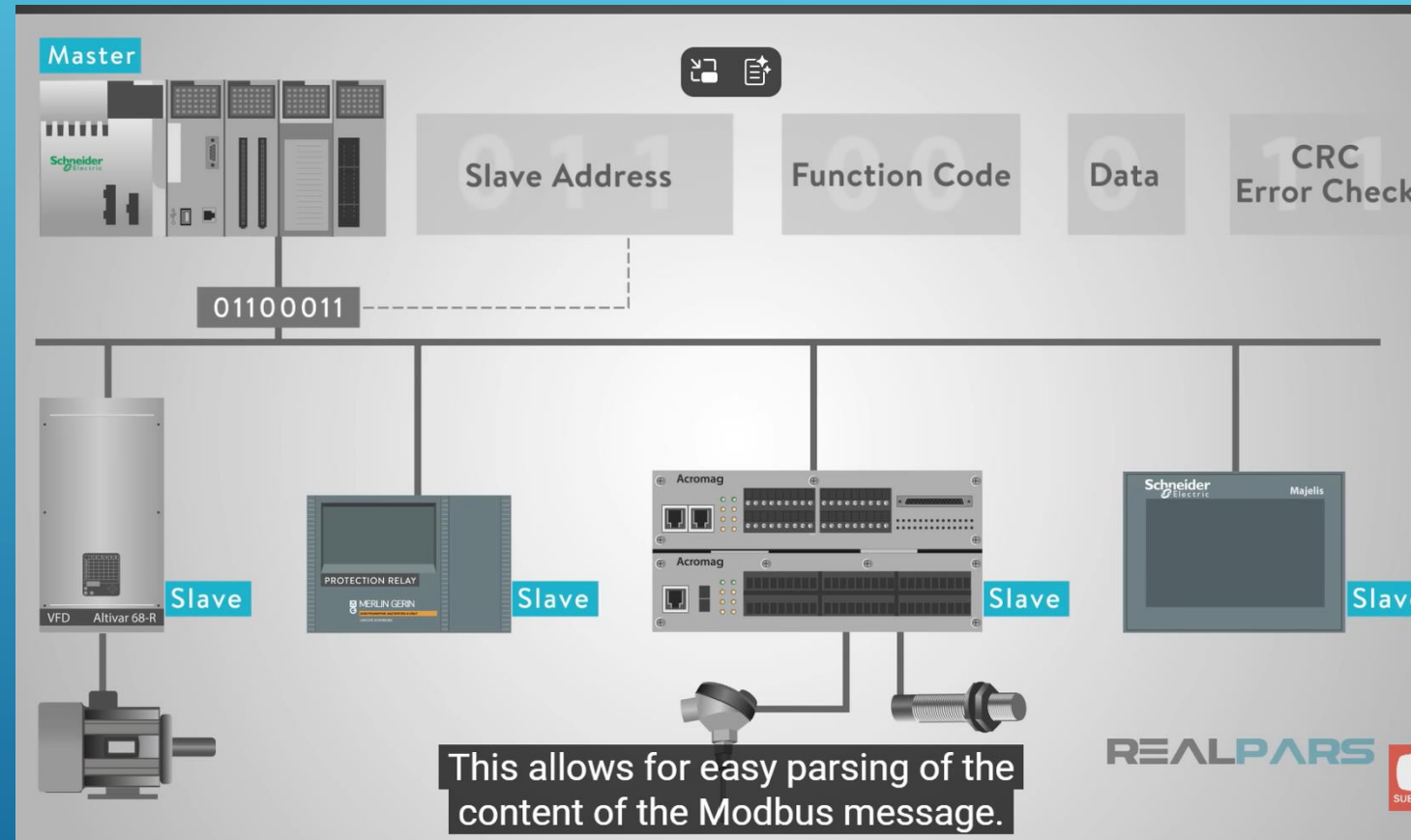
The PLC (Master) creates and sends a Modbus "frame" (via Modbus TCP or RTU). This frame must contain the following information:

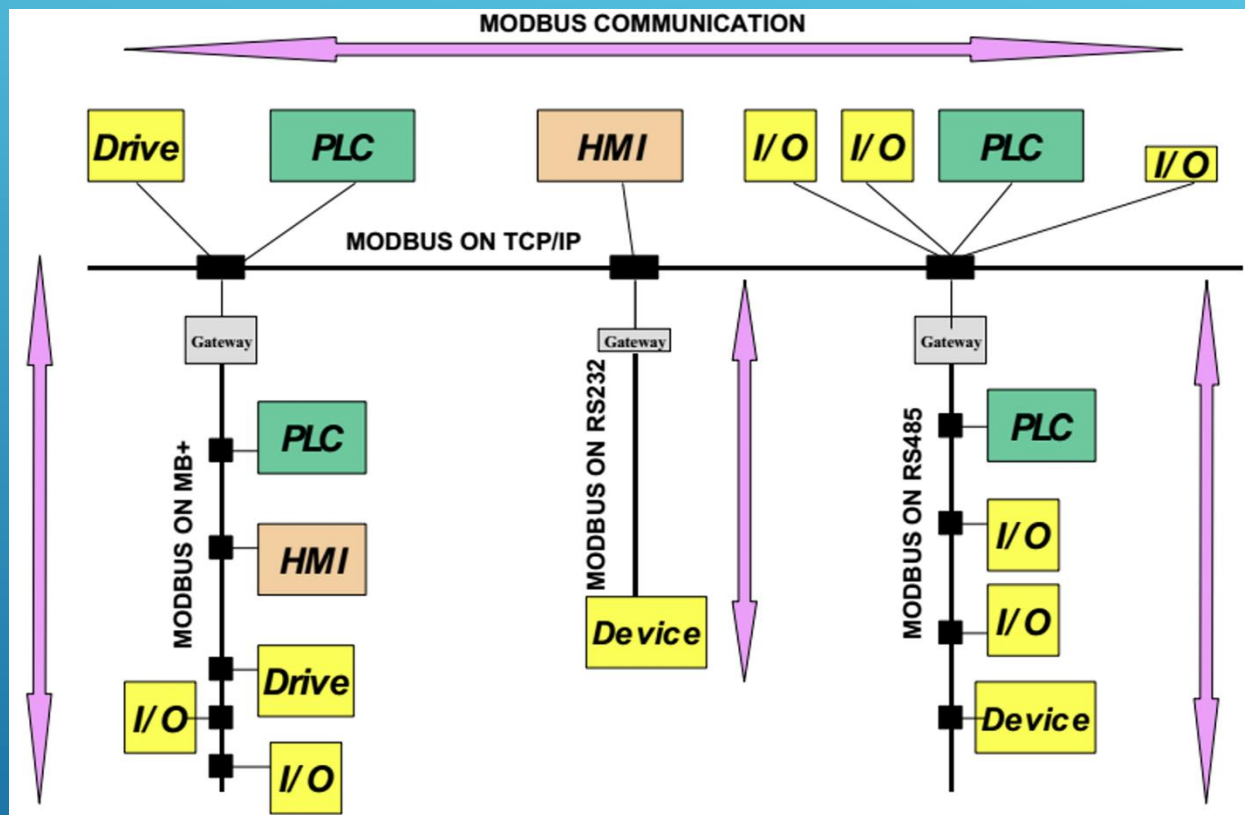
STEP 4: LINK TO MODBUSPROTOCOL.ORG

ADU AND PDU

<https://www.youtube.com/watch?v=JBGaInI-TG4>

[modbus TELEMECANIQUE - Recherche Vidéos](#)



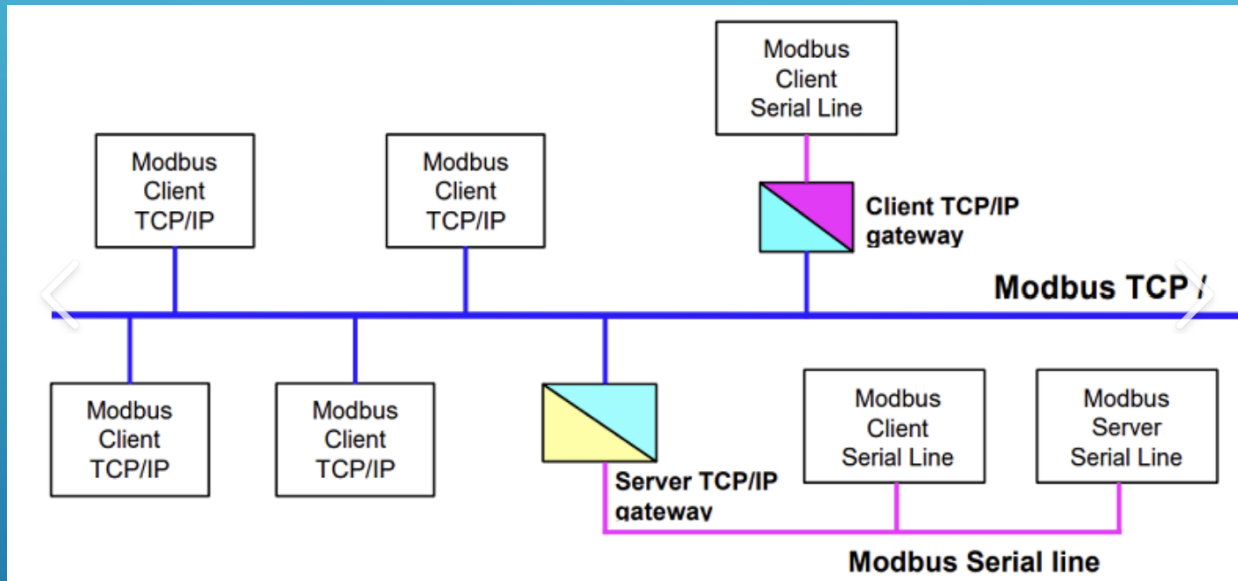


STEP 5: MODBUS TCP HOW TO ADAPT

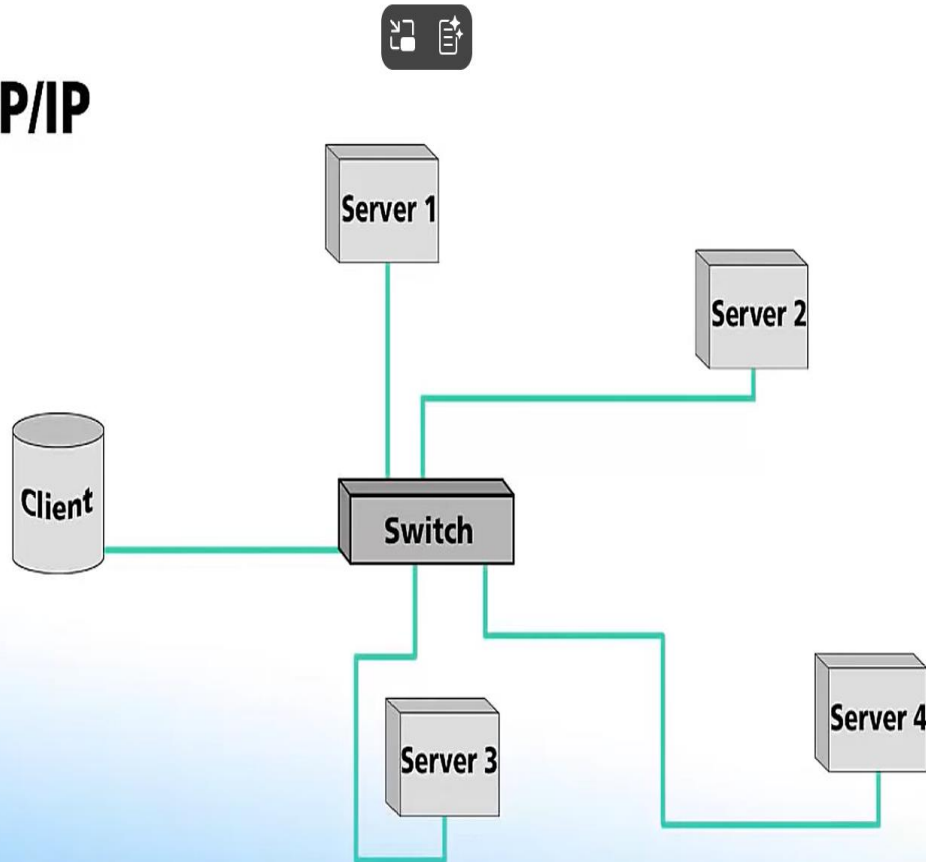
The necessity of Adding MBAP Header to support Protocol TCP\IP

STEP 5: MODBUS TCP THROUGH GATEWAYS

The necessity of Adding MBAP Header to support Protocol TCP\IP



TCP/IP

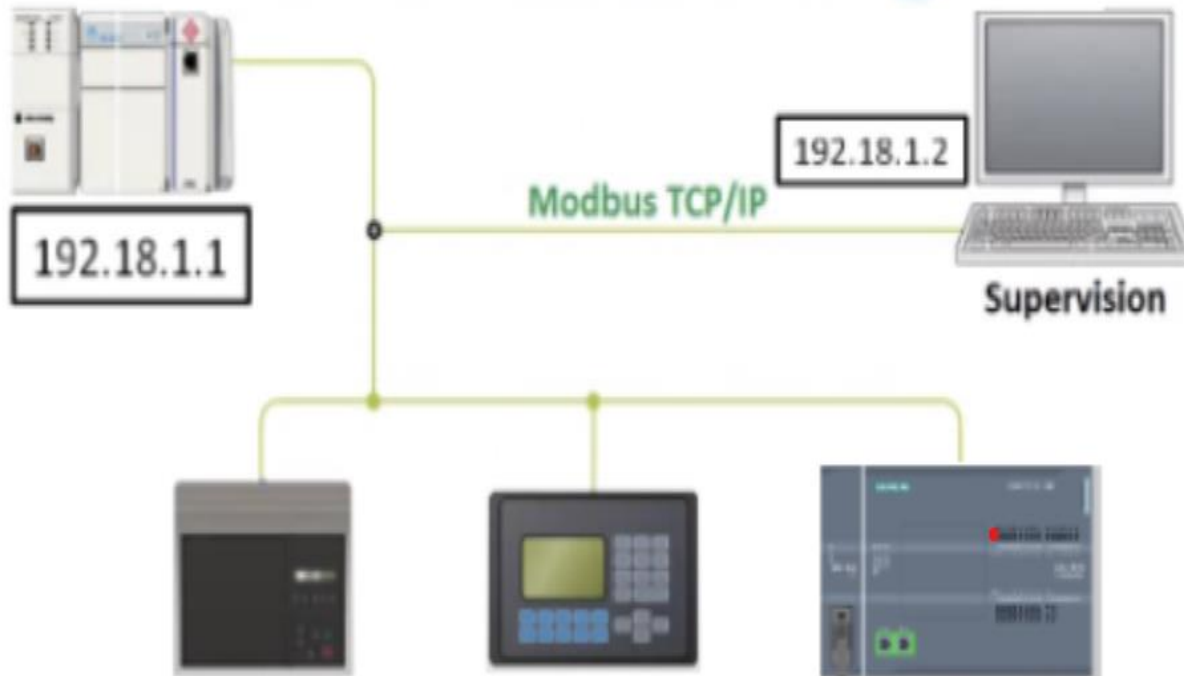


STEP 5: MODBUS TCP

The primary/replica request model is replaced by a client/server dialogue.

The necessity of Adding MBAP Header to support Protocol TCP\IP

Modbus TCP/IP



Modbus TCP là gì

STEP 5: MODBUS TCP

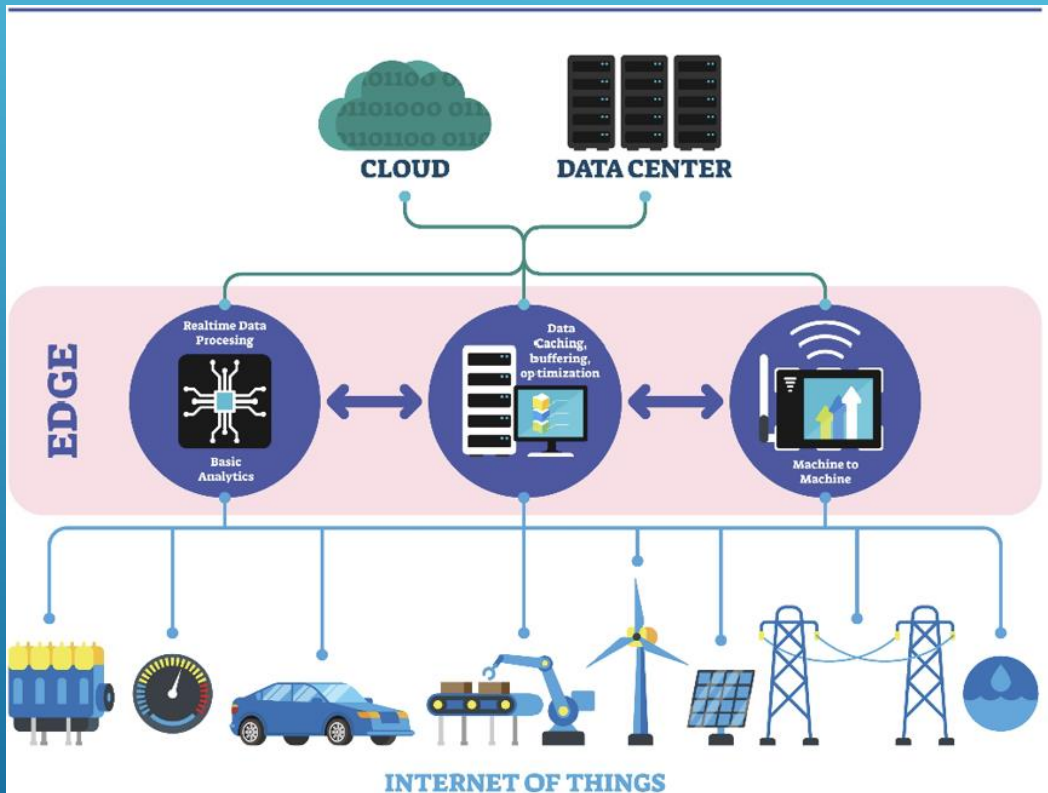
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The necessity of Adding MBAP Header to support Protocol TCP\IP

CONCLUSION: 3 KEY TALKING POINTS AND TWO OBJECTIVES

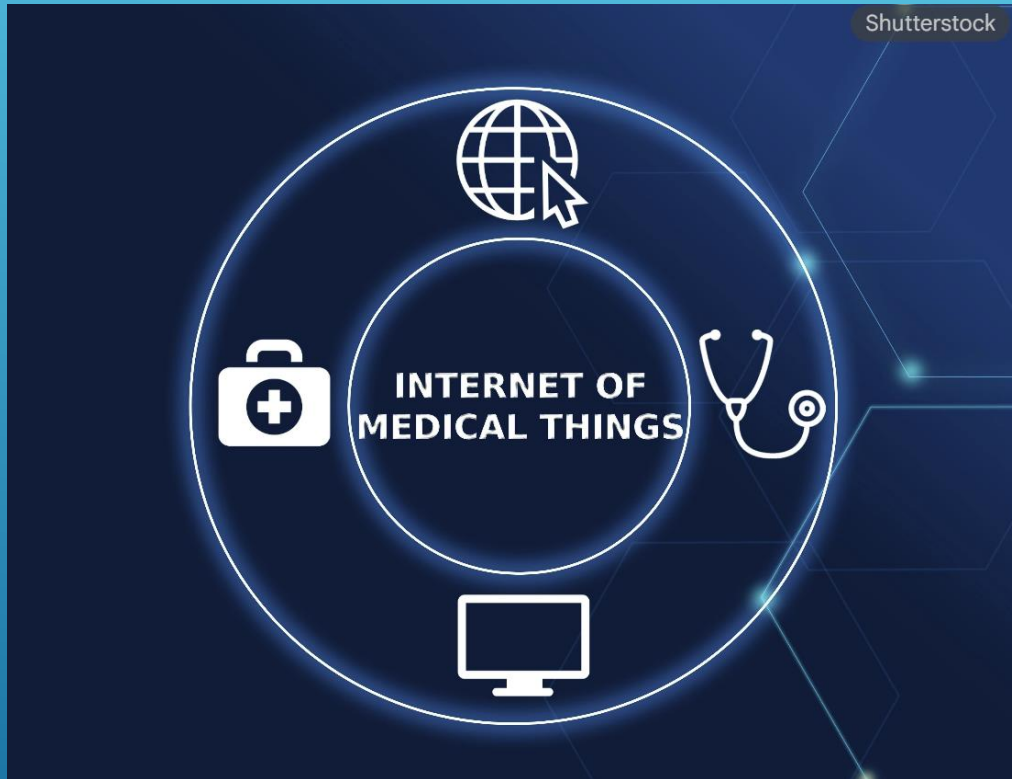
II. The Two Objectives

1. To Understand the crucial difference between Protocols
2. To Design a basic IIoT Architecture



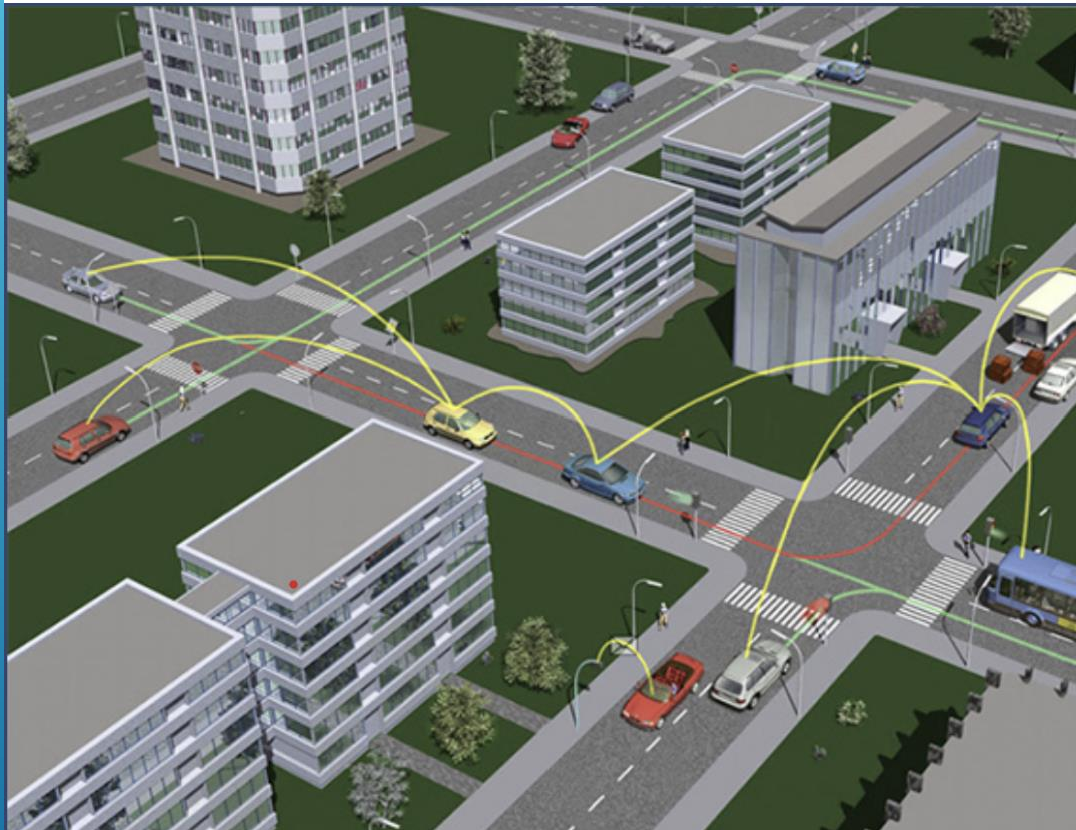
SECTION 4. BEYOND THE FACTORY: FUTURE IIOT ADAPTATION AND VERTICAL APPLICATIONS

This final section expands the discussion to new domains, illustrating how the core IIoT principles of protocol adaptation apply to other critical, data-intensive fields.



THE PROBLEM OF IOMT ADAPTATION

In a similar way that Modbus must adapt to IT, specialized fields like the **Internet of Medical Things (IOMT)** also face the problem of adapting their proprietary data streams and network requirements to standard, secure IT/Cloud infrastructure. These systems must also follow the architectural objectives of designing a basic IIoT architecture and understanding protocol differences.



VEHICULAR AD-HOC NETWORKS (VANET)

The principles of real-time communication and robust networking extend to the transportation sector.

<https://youtu.be/14fOqMBn9aw?t=28>

Agenda: IIoT IN 2025

I. SECTION 1: From the Smart Factory to Big Data Smart Factory Definition (OT/IT Convergence)

Big Data Challenge: The Volume and Velocity Challenge

The Link to Protocols (Bridging the OT/IT Gap)

Difference in Priorities (Safety vs. Security)

Difference in Data Context (Raw Value vs. Object Model)

The Role of the Bridge: OPC UA Protocol

II. SECTION 2: The Enduring Utility of Modbus Modbus Protocol: Definition and Historical Model (Master/Slave)

Q1: Why is Modbus still used in OT Controllers?

6.1. Simplicity and Ubiquity (Low Complexity, High Adoption)

6.2. Speed and Predictability (Deterministic Timing)

6.3. Ease of Diagnostics

III. SECTION 3: Let's Build a Scenario

Key Players and Platforms (Siemens MindSphere, AWS/Azure, Schneider Electric)

Q2: Who uses these protocols and why is Modbus needed?

Scenario: Modbus Data Request (Master to Slave)

Modbus TCP Adaptation: Replacing Master/Slave with Client/Server Dialogue (Necessity of MBAP Header)

IV. SECTION 4: Future IIoT Adaptation and Vertical Applications

The Problem of IOMT Adaptation (Internet of Medical Things)

Vehicular Ad-Hoc Networks (VANET)

V. Overall Objectives

To Understand the crucial difference between Protocols

To Design a basic IIoT Architecture